

Comparison between Evacuated Tube Collector & Flat Plate Collector

S.No	Evacuated Tube Collector	Flat Plate Collector
1	Quick heat generation	Slow heat generation
2	Collector efficiency on higher temperature is high	Collector efficiency on higher temperature is low.
3	Heat loss in the tubes during the daytime is negligible (evacuated tubes)	Heat loss in the collector & tank during the day-time is high due to convection.
4	Convection and Convecting losses is low.	Convection and Convecting losses is high.
5	Emissivity is low.	Emissivity is high.
6	Satisfactory performance even in extreme cold condition (-18 deg. C)	Freezing of water will take place at high altitude causing damage to the collector.
7	Temperature range from 60deg. to 120 deg.	Temperature range from 60 deg. to 80 deg.
8	System hot water tank only is insulated using polyurethane insulation material which does not absorb water or moisture	Collector & tank insulated with glasswool/ rockwool, absorbs moisture & gets wet during monsoons reducing the efficiency of the system
9	Negligible scaling of tubes which can be cleaned manually (inner tube dia. Is 37mm). Loss of efficiency consequently is minimal	Heavy scaling of the copper/aluminum tubes which cannot be cleaned manually as the bore dia. is 12.50 mm. giving rise to substantial loss in efficiency of the system.
10	The collector glass tube absorbers being cylindrical the incident sun's rays on the tubes is at 90 degrees throughout the day. Hence peak heat absorption always.	The collector fins & tubes being flat the incident sun's rays will be at 90 degrees at noon only for peak absorption
11	In locations with average availability of solar energy over-sizing of the system glass tube collectors is not required.	Higher system sizing is required to get the desired result. Hence added cost.
12	Heat exchanger not required	Heat exchanger required
13	Advanced technology at competitive prices that is System Cost per unit water is low.	Old technology at higher prices
14	Hot water availability for 350 days in a year.	Hot water availability for 300 days in a year claimed.
15	System life above 15 years.	System life above 15 years
16	It is very easy to replace glass tube.	Difficult and expensive to replace glass sheet
17	Water quality is not effect the system.	Water quality effect the heating system forming scale over metal tube.
18	It has low maintenance	Its required high maintenance.
19	Grouting of Collectors not required.	Grouting of collectors are required.

S.no	Closed Cycle Gas Turbine	Open Cycle Gas Turbine
1.	In this first the air is compressed in the compressor and then heated in a heating chamber. As the air is heated by an external source, so the amount of the gas remains same.	Here the compressed air is heated in the combustion chamber. So the products of combustion gets mixed with the heated air and hence the amount of gas does not remain same.
2.	In closed cycle gas turbine, the gas that comes out from the gas turbine passes into the cooling chamber.	In open cycle, the gases coming out from the gas turbine is exhausted in the atmosphere.
3.	The working fluid is circulated continuously.	The working fluid is replaced continuously.

4.	Any other fluid that possesses better thermodynamic properties like helium can be used.	Here only air can be used as the working fluid.
5.	No earlier wear of the turbine blades, because the enclosed gas does not gets contaminated while flowing through the heating chamber,	Earlier wear of turbine blades, as the air from the atmosphere enters the combustion chamber it gets contaminated.
6.	It is best suited for the stationary installation and marine uses because the air from the turbine is cooled by the circulating water.	It is best suited for the moving vehicle because the air from the turbine is discharged into the atmosphere.

7.	The maintenance cost of this type of turbine is high.	Its maintenance cost is low.
8.	More mass of installation per kW.	Less mass of installation per kW.

- *So, the Comparison of Gas Turbine with I. C. Engine is as below*

01) In winter nights, especially in cold regions, some provision has to be made to prevent freezing in internal combustion engines using **water cooling**.

This should be done when the temperature of the atmosphere falls to sub-zero values.

Anti-Freezer is generally used for this purpose.

There is no danger of freezing, as gas turbine does not use any water.

On the other hand, **gas turbine** works better if the temperature of the atmosphere falls below.

02) The output from a gas turbine increases if the inlet temperature of the working fluid decreases.

At the high altitudes where the temperature of the air is quite low, the output of the aviation gas turbine can also be well maintained.

While the output of **internal combustion engine** at high altitude would decrease.

03) For the same output, the weight per horsepower of a gas turbine is about one third that of a reciprocating **piston** type internal combustion engine.

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04) A low-grade fuel can be used in a gas turbine. In some cases, even **coal** and peat have been burnt.

Low-grade fuel cannot be used in internal combustion engines.

05) The fuel and **lubrication** cost in a gas turbine is lower compared to an internal combustion engine.

A draft tube is a pipe of gradually increasing area which connects the outlet of the runner to the tail race and is used to discharge water from the turbine exit to the tail race. One end of the draft tube is connected to the outlet of the runner while the other end is submerged below the level of water in tail race.

Functions:

1. It permits a negative head to be established to be established at the outlet of the runner and thereby increasing the NET HEAD on the turbine.
2. It converts a large proportion of kinetic energy which was being rejected at the outlet of the turbine into useful pressure energy.

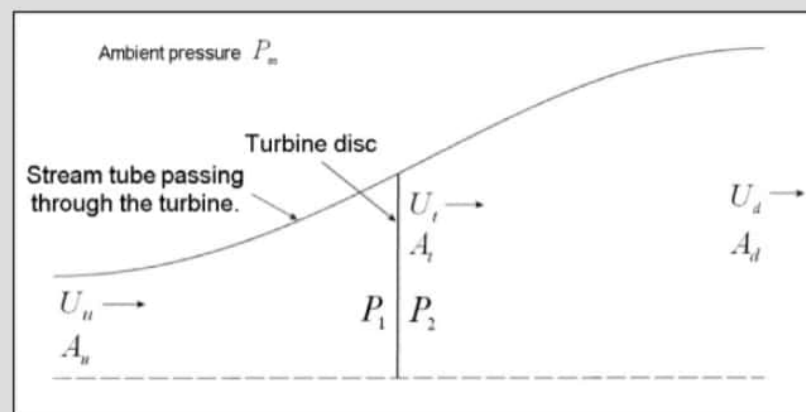
-> As a result of which the turbine develops more power and its efficiency also increases.

These are only used in Reaction Turbines, since the head is very low in such turbines.

17. The Betz limit - and the maximum efficiency for horizontal axis wind turbines.

Can it be exceeded and does it apply to vertical axis wind turbines?

The simplest model of a wind turbine is the so-called actuator disc model where the turbine is replaced by a circular disc through which the airstream flows with a velocity U_t and across which there is a pressure drop from P_1 to P_2 as shown in the sketch. At the outset, it is important to stress that the actuator disc theory is useful (as will be shown) in discussing overall efficiencies of turbines but it does not help at all with how to design the turbine blades to achieve a desired performance.



The power developed by the wind turbine is

$$\text{Power} = (P_1 - P_2) A_t U_t \quad (1)$$

where A_t is the turbine disc area. Volume flow continuity gives

$$A_u U_u = A_d U_d = A_t U_t \quad (2)$$

From momentum conservation, the force exerted on the turbine is equal to the momentum change between the flow far upstream of the disc to the flow far downstream of the disc. Thus

$$(P_1 - P_2) A_t = \text{Mass flow} \times \text{Velocity difference} = \rho A_u U_u (U_u - U_d) \quad (3)$$

The final basic equations are Bernoulli's equation applied upstream and downstream of the actuator disc

$$P_\infty + \frac{1}{2} \rho U_u^2 = P_1 + \frac{1}{2} \rho U_t^2 \quad (4a)$$

$$P_\infty + \frac{1}{2} \rho U_d^2 = P_2 + \frac{1}{2} \rho U_t^2 \quad (4b)$$

where P_∞ is the ambient pressure in the flow both far upstream and far downstream of the actuator disc.

From equations (4a), (4b), (3) and (2)

$$(P_1 - P_2) = \frac{1}{2} \rho (U_u^2 - U_d^2) = \rho \frac{A_u}{A_t} U_u (U_u - U_d) = \rho U_t (U_u - U_d) \quad (5)$$

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whence

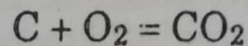
$$U_t = \frac{1}{2} (U_u + U_d) \quad (6)$$

i.e. the velocity through the actuator disc is the mean of the upstream and downstream velocities in the stream tube. Finally, from equations (1), (5) and (3), the efficiency is given by

$$\eta = \frac{\text{Power}}{\frac{1}{2} \rho A_t U_u^3} = \frac{1}{2} \left(1 - \frac{U_d}{U_u} \right) \left(1 + \frac{U_d}{U_u} \right)^2 \quad (7)$$

3.15 Calculation of Chimney Height

Air is required to burn coal or fuel in the furnace. We know



This shows that volume of chimney gases produced by the complete combustion of 1 kg of coal or oil is same as that of air required to support combustion if the temperature of flue gases and air is same.

Let H = Height of chimney above grate level in metres.

W = Weight of air in kg required per kg of fuel.

T = Average absolute temperature of chimney gases in °K.

T_1 = Absolute temperature of air outside the chimney in °K.

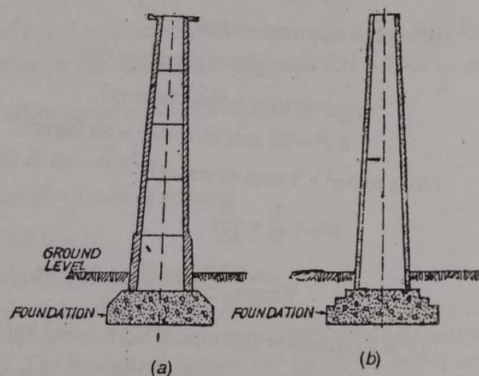


Fig. 3.36.

Weight of chimney gases produced

$$= (W + 1) \text{ kg of fuel burnt}$$

Volume of chimney gases at 273°K

$$= \text{volume of 1 kg air at } 273^\circ\text{K}$$

$$= \frac{RT}{P} = \frac{29.27 \times 273}{1.03 \times 10^4} \times 0.7734 \text{ m}^3.$$

Therefore, volume of one kg of air at $T_1^\circ\text{K}$

$$= 0.7734 \times \frac{T_1}{273}$$

Thus volume of W kg of air at $T_1^\circ\text{K}$

$$= \frac{0.7734 \times T_1 \times W}{273}$$

Density of air at $T_1^\circ\text{K} = \frac{\text{Mass}}{\text{Volume}}$

$$= W \times \frac{273}{0.7734 \times T_1 \times W} = \frac{273}{0.7734 \times T_1}$$

Therefore, pressure at the grate due to a column of cool air of H metres height

$$= \frac{273}{0.7734 \times T_1} \times H = 1.293 \times \frac{273}{T_1} \times H \text{ kg/m}^2 \dots (i)$$

Volume of chimney gases at $T^\circ\text{K}$ per kg of air

$$= 0.7734 \times \frac{T}{273}$$

Volume of $(W + 1)$ kg of chimney gases at $T^\circ\text{K}$

$$= 0.7734 \times \frac{T}{273} \times W$$

Density of chimney gases at $T^\circ\text{K}$

$$= \frac{\text{Mass}}{\text{Volume}} = (W + 1) \times \frac{273}{0.7734 \times T \times W}$$

$$= 1.293 \left(\frac{W + 1}{W} \right) \times \frac{273}{T}$$

$$= 353 \left[\frac{W + 1}{W} \right] \times \frac{1}{T}$$

Therefore, pressure at the grate by a column of hot gases of height H metre = Density $\times H$ kg/m²

$$= 1.293 \left(\frac{W + 1}{W} \right) \times \frac{273}{T} \times H \text{ kg/m}^2 \dots (ii)$$

Therefore, pressure P causing the draught

$$P = \left[\left(1.293 \times \frac{273 \times H}{T_1} \right) - \left(1.293 \times \frac{(W + 1)}{W} \times \frac{273}{T} \times H \right) \right]$$

$$P = 353 H \left[\frac{1}{T_1} - \left(\frac{W + 1}{W} \right) \times \frac{1}{T} \right] \text{ kg/m}^2.$$

If this draught is h mm of water as measured by a U-tube water manometer.

$$h = 353 H \left[\frac{1}{T_1} - \left(\frac{W + 1}{W} \right) \times \frac{1}{T} \right] \text{ mm of water}$$

(We know $1 \text{ kg/m}^2 = 1 \text{ mm of water column}$)

Let h' = height of column of hot gases.

$$h' \times 1.293 \times \left(\frac{W + 1}{W} \right) \times \frac{273}{T} = P$$

$$= 353 H \left[\frac{1}{T_1} - \left(\frac{W + 1}{W} \right) \times \frac{1}{T} \right]$$

POWER

$$h' = \frac{353 H \left[\frac{1}{T_1} - \left(\frac{W+1}{W} \right) \times \frac{1}{T} \right]}{1.293 \times 273 \times \left(\frac{W+1}{W} \right) \times \frac{1}{T}}$$

$$h' = H \left[\left(\frac{W}{W+1} \right) \times \frac{T}{T_1} - 1 \right]$$

Example 4.14 A boiler uses 2000 kg/h of coal. The temperature of air supplied is 300 K, and the average temperature of the flue gas leaving the chimney is 650 K. The 33-m high steel chimney produces a draught of 20 mm of water column. Determine the (a) quantity of air supplied per kg of coal, (b) the draught in terms of column of hot gases, and (c) the base diameter of the chimney, assuming that 10% of the theoretical draught is used for creating the flow velocity of gases through the chimney.

Solution

$$(a) \quad h = 353 H \left[\frac{1}{T_a} - \frac{m+1}{m} \frac{1}{T_g} \right]$$

$$20 = 353 \times 33 \left[\frac{1}{300} - \frac{m+1}{m} \frac{1}{650} \right]$$

$$m = 19.727 \text{ kg air/kg coal}$$

Ans. (a)

(b) Let H_g be the height of the hot gas column producing the draught. Then

$$\Delta p = \rho_g g H_g = 353 g H \left[\frac{1}{T_g} - \frac{m+1}{m} \frac{1}{T_a} \right]$$

where

$$\rho_g = 353 \frac{m+1}{m} \frac{1}{T_g} = 353 \times \frac{20.727}{19.727} \times \frac{1}{650} = 0.5706 \frac{\text{kg}}{\text{m}^3}$$

$$H_g = H \left[\frac{m}{m+1} \frac{T_g}{T_a} - 1 \right] = 33 \left[\frac{19.73}{20.73} \times \frac{650}{300} - 1 \right] = 35 \text{ m} \checkmark$$

Ans. (b)

$$\begin{aligned} \text{(c) Mass flow of flue gas} &= \text{mass of coal} + \text{mass of air} \\ &= 2000 (1 + 19.727) / 3600 \\ &= 11.515 \text{ kg/s} \end{aligned}$$

$$\therefore \text{Volume flow of flue gas} = \frac{11.515}{0.5706} = 20.18 \text{ m}^3/\text{s}$$

$$\begin{aligned} \text{Flue gas velocity, } V_g &= \sqrt{2gH_g} = \sqrt{2 \times 9.81 \times 0.1 \times 35} \\ &= 8.29 \text{ m/s} \checkmark \end{aligned}$$

$$\frac{\pi}{4} d^2 \times V_g = 20.18$$

$$\frac{\pi}{4} d^2 = \frac{20.18}{8.29}$$

$$d = \text{base diameter of chimney}$$

$$= 1.76 \text{ m}$$

Ans. (c)